CDA 106
Numerical Methods for Data Science

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Consider a given matrix $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

Assume that

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \quad \& \quad U = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

If $A = LU$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ & \ldots & \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ \underline{\qquad\qquad} & & \\ \underline{\qquad\qquad} & & \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ \underline{\qquad\qquad} & & \\ \underline{\qquad\qquad} & & \end{bmatrix}$$

or $\underbrace{\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}}_{\text{known}} = \underbrace{\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}}_{\text{unknown}} \underbrace{\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}}_{\text{unknown}}$ 12 unknown

From here, by taking simple product of RHS (L & U matrices) and comparing with LHS we get

① $a_{11} = l_{11} u_{11}$

② $a_{12} = l_{11} u_{12}$

③ $a_{13} = l_{11} u_{13}$

④ $a_{21} = l_{21} u_{11}$

⑤ $a_{22} = l_{21} u_{12} + l_{22} u_{22}$

⑥ $a_{23} = l_{21} u_{13} + l_{22} u_{23}$

⑦ $a_{31} = l_{31} u_{11}$

⑧ $a_{32} = l_{31} u_{12} + l_{32} u_{22}$

⑨ $a_{33} = l_{31} u_{13} + l_{32} u_{23} + l_{33} u_{33}$

- Here, l_{ij} & u_{ij} are unknown and a_{ij} are known
- Here, we have 9 eqns in 12 unknowns

Therefore, in order to get solution we can fix $12 - 9 = 3$ unknowns

We can solve above eq^n ①-⑨, by fixing.

(A) $l_{11} = l_{22} = l_{33} = 1$ [Do little LU decomposition]

$u_{11} = a_{11}$, $u_{12} = a_{12}$, $u_{13} = a_{13}$, $l_{21} = \dfrac{a_{21}}{u_{11}}$, $l_{31} = \dfrac{a_{31}}{u_{11}}$ ✓

$u_{22} = \dfrac{a_{22} - l_{21} u_{12}}{l_{22}}$, $u_{23} = \dfrac{a_{23} - l_{21} u_{13}}{l_{22}}$ ✓

$l_{32} = \dfrac{a_{32} - l_{31} u_{13}}{u_{22}}$, $u_{33} = \dfrac{a_{33} - l_{31} u_{13} - l_{32} u_{23}}{l_{33}}$ ✓

(B) $u_{11} = u_{22} = u_{33} = 1$ [Crout's LU decomposition]

Similarly as in case (A) we can find l_{ij} & u_{jj} in case B.

Example Consider the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 3 & 5 & 3 \end{bmatrix}$ and find the LU decomposition of A.

Ans Suppose we try Crout's decomposition

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 3 & 5 & 3 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

Along first row:

$1 = l_{11}$, $l_{11} u_{12} = 1$, $l_{11} u_{13} + 0 = 1$

$\Rightarrow \boxed{l_{11} = 1}$ $\boxed{u_{12} = 1}$ $\boxed{u_{13} = 1}$

Along the first column: $l_{11} = 1$ $\boxed{4 = l_{21}}$ $\boxed{3 = l_{31}}$

$3 = l_{21} u_{12} + l_{22} \Rightarrow \boxed{l_{22} = -1}$ $\quad -1 = l_{21} u_{13} + l_{22} u_{23} \Rightarrow u_{23} = 4 + 1 = 5$

$5 = l_{31} u_{12} + l_{32} \Rightarrow \boxed{l_{32} = 2}$ $\quad 3 = l_{31} u_{13} + l_{32} u_{23} + l_{33} \Rightarrow \boxed{l_{33} = -10}$

Therefore, we get

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 4 & 3 & -1 \\ 3 & 5 & 3 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 4 & -1 & 0 \\ 3 & 2 & -10 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{bmatrix}}_{U}$$

These were the same L & U matrices, I gave you in the example in last class, while solving system of eqn $AX = \begin{bmatrix} 1 \\ 6 \\ 4 \end{bmatrix}$

<u>Exercise</u>:- Find LU decomposition using Doolittle method for above A.
(left as ~~a~~ homework)

Example Use Crout's method to find LU decomposition of

$$A = \begin{bmatrix} 3 & 2 & -3 \\ 2 & 2 & 5 \\ 1 & 1 & -1 \end{bmatrix}.$$

Ans

$$\underbrace{\begin{bmatrix} 3 & 2 & -3 \\ 2 & 2 & 5 \\ 1 & 1 & -1 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}}_{U}$$

Do the calculations to get

$$\underbrace{\begin{bmatrix} 3 & 2 & -3 \\ 2 & 2 & 5 \\ 1 & 1 & -1 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} 3 & 0 & 0 \\ 2 & 2/3 & 0 \\ 1 & 1/3 & -7/2 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 1 & 2/3 & -1 \\ 0 & 1 & 21/2 \\ 0 & 0 & 1 \end{bmatrix}}_{U}.$$

Now consider the following matrix

$$\underline{\underline{A}} = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 2 & 5 \\ 3 & 2 & -3 \end{bmatrix}$$

$AX = b$, for some fix b, will have same sol^n in both this case & A in previous slide.

This new A matrix, we found by changing $R_1 \leftrightarrow R_3$ in previous A.

However, if we look at the LU decomposition, then we have

$$\overset{A}{\begin{bmatrix} 1 & 1 & -1 \\ 2 & \underset{a_{22}}{2} & \underset{a_{23}}{5} \\ 3 & 2 & -3 \end{bmatrix}} = \overset{L}{\begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}} \overset{U}{\begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}}$$

[using Crout's method]

$l_{11} = 1$, $u_{12} = 1$, $u_{13} = -1$, $l_{21} = 2$, $l_{31} = 3$

$2 = l_{21} u_{12} + l_{22} \Rightarrow \boxed{l_{22} = 0}$

$5 = l_{21} u_{13} + l_{22} u_{23}$ ✗ you cannot solve u_{23} from here, if $l_{22} = 0$

$\longrightarrow$ Then A does not have LU decomposition.

Try Do little

$$\begin{bmatrix} 1 & 1 & -1 \\ 2 & 2 & 5 \\ 3 & 2 & -3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$u_{11} = 1$, $\underline{u_{12} = 1}$, $u_{13} = -1$ $\quad \underline{l_{21} = 2}$ $\quad l_{22} = 3$

$$2 = \underline{l_{21}}\,\underline{u_{12}} + u_{22} \Rightarrow \boxed{u_{22} = 0}$$

So, LU decomposition is NOT possible.

Comments (Imp)

① For any $n \times n$ matrix $\underline{\underline{A}}$, in LU decomposition we get $\underbrace{n^2 \text{ eq}^n}_{\text{Size of } A}$ in $\underbrace{n^2+n \text{ variables}}_{\text{Sum of unknown in L \& U matrices}}$

② In case of L matrix

$$L_{n\times n} = \begin{bmatrix} l_{11} & 0 & 0 & \cdots & 0 \\ l_{21} & l_{22} & 0 & \cdots & 0 \\ l_{31} & l_{32} & l_{33} & \cdots & 0 \\ \vdots & & & & \\ l_{n1} & l_{n2} & l_{n3} & \cdots & l_{nn} \end{bmatrix} \begin{array}{l} \leftarrow 1 \text{ unknown} \\ \leftarrow 2 \text{ unknown} \\ \vdots \\ \leftarrow n \text{ unknown} \end{array}$$

Total unknown $= 1+2+3+\cdots+n = \sum_{k=1}^{n} k = \frac{n(n+1)}{2} = \frac{n^2+n}{2}$ ✓

③ Similarly, the upper triangular matrix $\underline{\underline{U_{n\times n}}}$, will also have $\frac{n^2+n}{2}$ unknowns ✓

(4) Therefore, in order to solve n^2 eq^n in n^2+n unknowns, we must fix 'n' unknowns so that we get n^2 eq^{ns} in n^2 unknowns.

This can be done in one of the following special ways.

- Fix l_{ii} $\forall i$ (The method is called Do little)
- Fix u_{ii} $\forall i$ (The method is called Crout's)

(5) In case any of the $l_{ii}=0$ (OR $u_{ii}=0$) for some i then the LU decomposition by above process fails.

Question: Can we say, apriori, whether a given matrix has LU decomposition or not.

YES!

Result:

Any $n \times n$ matrix A has LU decomposition if and only if all the leading principle minors are non-zero.

Principle minors

$$A = \begin{bmatrix} \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \\ \times & \times & \times & \times & \times \end{bmatrix}$$

let us consider our earlier example

1

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 2 & 5 \\ 3 & 2 & -3 \end{bmatrix}$$

$|1| = 1$, $\begin{vmatrix} 1 & 1 \\ 2 & 2 \end{vmatrix} = 0$ $\begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 5 \\ 3 & 2 & -3 \end{vmatrix} = 1(-6-10) - 1(-6-15) + 1(4-6)$

$= -16 + 21 - 2 = \underline{\underline{3}}$

2

Further, for $A = \begin{bmatrix} 3 & 2 & -3 \\ 2 & 2 & 5 \\ 1 & 1 & -1 \end{bmatrix}$

$|3| = 3$, $\begin{vmatrix} 3 & 2 \\ 2 & 2 \end{vmatrix} = 2$ $\begin{vmatrix} 3 & 2 & -3 \\ 2 & 2 & 5 \\ 1 & 1 & -1 \end{vmatrix} = 3(-2-5) - 2(-2-5) - 3(0)$

$= -21 + 14 = \underline{\underline{-7}}$

If and only if (Necessary & sufficient)

Statement A $\Rightarrow$ Statement B

then A is sufficient for B

Statement B is considered as necessary.

And statement B $\Rightarrow$ Statement A.

Diff $\Rightarrow$ Cont.

① Diff suff. Cont.

② Cont necessary Diff.